When Geometry Meets Causality:
Statistical Methods for Cosmic Web Analysis and Beyond

Yikun Zhang
Department of Statistics, University of Chicago · NSF–Simons AI Institute for the Sky (SkAI Institute) · yikunz@uchicago.edu

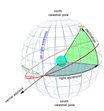
Funded by the U.S. National Science Foundation and the Simons Foundation

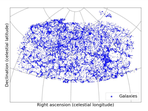
Three Statistical Challenges Behind Cosmic Web Analysis

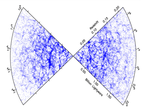
1. Nonlinear Geometry and Low-Dimensional Structures: $\Omega_q = \{\boldsymbol{x} \in \mathbb{R}^{q+1} : ||\boldsymbol{x}||_2 = 1\}$.

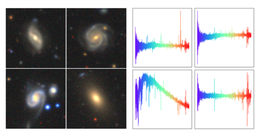
2. High Dimensionality and Missingness: $\boldsymbol{X} \in \mathbb{R}^d$, $d \gg n$, and $Y \in \mathbb{R} \cup \{\mathrm{NA}\}$.

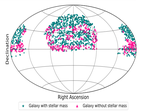
3. Confounding and Positivity Violations: $p_{T|\boldsymbol{X}}(t \mid \boldsymbol{x}) = 0$ for some $(t, \boldsymbol{x}) \in \mathcal{T} \times \mathcal{X}$.

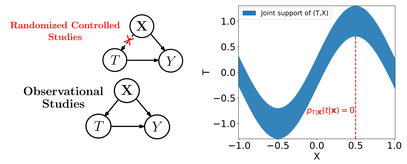

Celestial sphere Ω_2.


Galaxies on Ω_2 (fixed redshift).


Galaxies across redshifts $\Omega_2 \times \mathbb{R}$.


Photometric and spectral features.

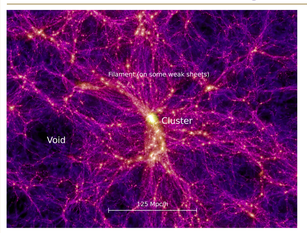

Galaxies in a high redshift slice.



Main Theme: Develop **geometry-aware** statistical learning and causal inference methods to address these challenges.

Geometry-Aware Cosmic Web Detection

Background and Data



Cosmic Web: The large-scale network structure revealing the non-uniformity of matter distribution in the Universe.

- 0D **galaxy clusters** (*cosmic nodes*);
- 1D **filaments** connecting clusters;
- 2D tenuous **sheets** or **walls**;
- 3D nearly empty **cosmic voids**.

We focus on **cosmic nodes and filaments**: key tracers of matter concentration, galaxy evolution, and weak lensing.

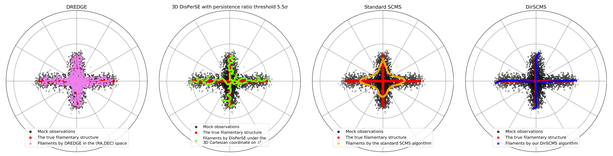
Observed Galaxy Data: Angular coordinates and redshifts $\{(\alpha_i, \delta_i, z_i)\}_{i=1}^n \subset \Omega_2 \times \mathbb{R}^+$.

- $\alpha_i \in [0°, 360°)$: *right ascension* (RA; celestial longitude);
- $\delta_i \in [-90°, 90°]$: *declination* (DEC; celestial latitude);
- $z_i \geq 0$: *redshift*, which provides distance information.

Existing Approaches and Limitations

3D Approach: Convert redshifts to comoving distances and detect filaments in 3D.
- Requires a cosmological distance model and is sensitive to "Finger-of-God" distortions.

2D Approach: Divide the survey by redshift and detect filaments within each 2D slice.
- Existing methods rely on a flat-sky approximation and neglect cosmic expansion.



Our method supports both filament detection on 2D spherical slices and an extension to full 3D light-cone analysis.

Proposed Model, Estimation, and Algorithm

Step 1 (Redshift Slices): Partition the Sloan Digital Sky Survey (SDSS-IV) into 325 non-overlapping slices of comoving thickness $\Delta L = 20\,\mathrm{Mpc}$ (*tomographic analysis*).

Step 2 (Survey Regions): Retain the selected *North Galactic Cap* and *South Galactic Cap* regions in each slice.

Step 3 (Galaxy Density): Map each galaxy's angular coordinates $(\alpha_i, \delta_i) \in [0, 360°) \times [-90°, 90°], i = 1, ..., n$ to Ω_2:

$$\boldsymbol{X}_i = (\cos\delta_i \cos\alpha_i, \cos\delta_i \sin\alpha_i, \sin\delta_i)\,, \quad i = 1, \ldots, n.$$

Fit a directional kernel density estimate (DirKDE):

$$f(\boldsymbol{x}) = \frac{C(b)}{n} \sum_{i=1}^n \exp\left(\frac{\boldsymbol{x}^T \boldsymbol{X}_i}{b^2}\right),$$

where $b > 0$ is the bandwidth and $C(b) > 0$ normalizes the density.

Step 4 (Filament Model): Represent the 1D cosmic filaments on Ω_2 as directional density ridges of f:

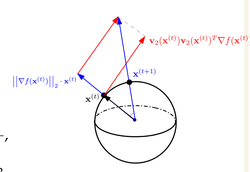
$$R_1(f) = \left\{\boldsymbol{x} \in \Omega_2 : \boldsymbol{v}_2(\boldsymbol{x})^T \operatorname{grad} f(\boldsymbol{x}) = 0, \lambda_2(\boldsymbol{x}) < 0\right\},$$

where $\operatorname{grad} f(\boldsymbol{x})$ is the *Riemannian gradient* of f at $\boldsymbol{x} \in \Omega_2$, and $\boldsymbol{v}_2(\boldsymbol{x})$ is the tangent-space eigenvector of the *Riemannian Hessian* $\mathcal{H}f(\boldsymbol{x})$ corresponding to its smaller eigenvalue $\lambda_2(\boldsymbol{x})$.

Step 5 (SCONCE Algorithm): Compute $R_1(f)$ using directional subspace constrained mean shift (DirSCMS), implemented in our filament finder `SCONCE`, until convergence ($t = 0, 1, \ldots$):

$$\tilde{\boldsymbol{x}}^{(t+1)} \leftarrow \boldsymbol{x}^{(t)} + \boldsymbol{v}_2(\boldsymbol{x}^{(t)})\boldsymbol{v}_2(\boldsymbol{x}^{(t)})^T \frac{\sum_{i=1}^n \boldsymbol{X}_i \exp\left(\frac{\boldsymbol{X}_i^T \boldsymbol{x}^{(t)}}{b^2}\right)}{\left|\left|\sum_{i=1}^n \boldsymbol{X}_i \exp\left(\frac{\boldsymbol{X}_i^T \boldsymbol{x}^{(t)}}{b^2}\right)\right|\right|_2},$$

$$\boldsymbol{x}^{(t+1)} \leftarrow \frac{\tilde{\boldsymbol{x}}^{(t+1)}}{\left|\left|\tilde{\boldsymbol{x}}^{(t+1)}\right|\right|}.$$

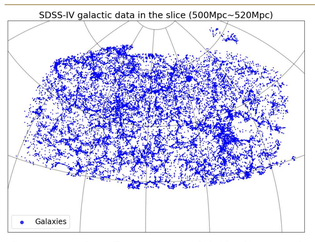


- Python package: `https://pypi.org/project/sconce-scms/`.

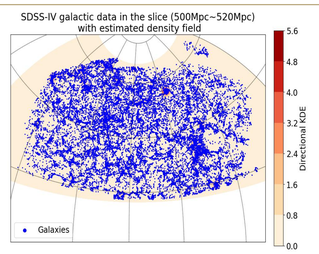
Step 6 (Uncertainty): Quantify the local filament uncertainties by nonparametric bootstrap.

Step 7 (Nodes and Filament Knots): Identify local modes of f using directional mean shift algorithm, and detect filament intersections via metric graph reconstruction.

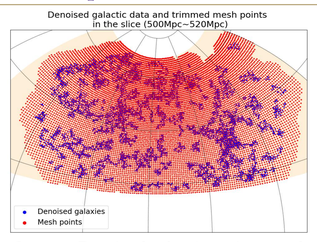
Cosmic Web Detection in an Example Redshift Slice



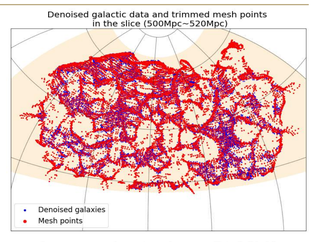
Steps 1 and 2: Construct redshift slices and select the North Galactic Cap.



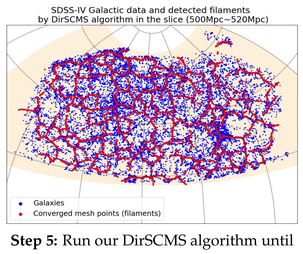
Step 3-1: Estimate the galaxy density field.



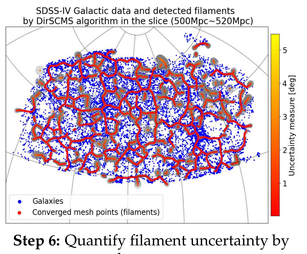
Step 3-2: Denoise the density estimate and initialize mesh points.



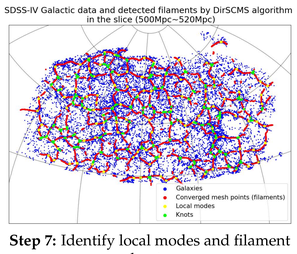
Steps 4 and 5: Apply our DirSCMS algorithm to mesh points (iteration 1).



Step 5: Run our DirSCMS algorithm until convergence.



Step 6: Quantify filament uncertainty by bootstrap.



Step 7: Identify local modes and filament knots.



Full catalog: Publicly available at `https://doi.org/10.5281/zenodo.6244866`.

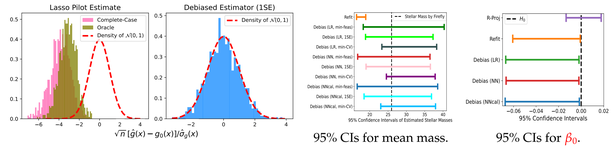
High-Dimensional Inference on the Cosmic Web

Question: Are galaxies farther from filaments *significantly less massive* after adjustment for observed features?

Challenges and Assumptions

1. Missing Outcomes: Stellar mass is unobserved for some galaxies.
- *Missing at random:* $Y \perp\!\!\!\perp R \mid (D_{\mathrm{fila}}, \boldsymbol{X})$, where $R = 1$ if Y is observed and $R = 0$ otherwise.

2. High-dimensional Covariates: Each galaxy has many observed features and nonlinear transformations.
- *Sparse linear model:* $Y = \beta_0 D_{\mathrm{fila}} + \boldsymbol{\beta}_1^T \boldsymbol{X} + \epsilon$, with $||\boldsymbol{\beta}_1||_0 = s_\beta \ll d$, $\mathbb{E}(\epsilon \mid D_{\mathrm{fila}}, \boldsymbol{X}) = 0$, and $\mathrm{Var}(\epsilon \mid D_{\mathrm{fila}}, \boldsymbol{X}) = \sigma_\epsilon^2$.



Finding: The proposed 95% confidence intervals (CIs) for β_0 lie below zero: galaxies farther from nearby filaments tend to have lower stellar masses after adjustment.

Debiased Inference Procedure

Write $\boldsymbol{Z}_i = (D_{\mathrm{fila},i}, \boldsymbol{X}_i^T)^T$, $\boldsymbol{z} = (d_{\mathrm{fila}}, \boldsymbol{x}^T)^T$, and $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1^T)^T \in \mathbb{R}^{d+1}$.

Step 1: Compute the **Lasso pilot estimate** with complete-case data:

$$\widehat{\boldsymbol{\beta}} \in \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^{d+1}} \left\{\frac{1}{n}\sum_{i=1}^n R_i (Y_i - \boldsymbol{Z}_i^T \boldsymbol{\beta})^2 + \lambda\, ||\boldsymbol{\beta}_1||_1\right\}, \qquad \lambda > 0.$$

Step 2: Obtain **consistent estimates of the observation probabilities** $\widehat{\pi}_i = \widehat{\mathbb{P}}(R_i = 1 | \boldsymbol{Z}_i)$ for $i = 1, ..., n$ by any machine learning method fitted to $\{(\boldsymbol{Z}_i, R_i)\}_{i=1}^n$.

Step 3: Solve the convex **debiasing program**:

$$\widehat{\boldsymbol{w}}(\boldsymbol{z}) \in \arg\min_{\boldsymbol{w} \in \mathbb{R}^n} \left\{\sum_{i=1}^n \widehat{\pi}_i \widehat{w}_i^2 : \left|\left|\boldsymbol{z} - \frac{1}{\sqrt{n}}\sum_{i=1}^n w_i \widehat{\pi}_i \boldsymbol{Z}_i\right|\right|_\infty \leq \frac{\gamma}{n}\right\}, \qquad \gamma > 0.$$

Step 4: Construct the **debiased estimator** of $g_0(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{\beta}$ as:

$$\widehat{g}^{\mathrm{debias}}(\boldsymbol{z}) = \boldsymbol{z}^T \widehat{\boldsymbol{\beta}} + \frac{1}{\sqrt{n}}\sum_{i=1}^n \widehat{w}_i(\boldsymbol{z}) R_i (Y_i - \boldsymbol{Z}_i^T \widehat{\boldsymbol{\beta}}).$$

Taking $\boldsymbol{z} = \boldsymbol{e}_1 = (1, 0, \ldots, 0)^T \in \mathbb{R}^{d+1}$ targets β_0.

Theory: Under regularity conditions, if $\sigma_g^2(\boldsymbol{z}) = \lim_{n\to\infty} \sigma_\epsilon^2 \boldsymbol{z}^T [\mathbb{E}(R\boldsymbol{Z}\boldsymbol{Z}^T)]^{-1} \boldsymbol{z} \in (0, \infty)$ exists,

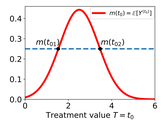
$$\sqrt{n}\left[\widehat{g}^{\mathrm{debias}}(\boldsymbol{z}) - g_0(\boldsymbol{z})\right] \xrightarrow{d} \mathcal{N}(0, \sigma_g^2(\boldsymbol{z})).$$

Causal Inference for Continuous Treatments Under Positivity Violations

Causal Setup and Estimand

- *Treatment* $T \in \mathcal{T} \subset \mathbb{R}$: Local environmental density of a galaxy.
- *Covariates* $\boldsymbol{X} \in \mathcal{X} \subset \mathbb{R}^d$: Observed treatment–outcome confounders.
- *Outcome* $Y \in \mathcal{Y} \subset \mathbb{R}$: Stellar mass or another galaxy property.
- *Potential Outcome* $Y^{(t_0)} \in \mathcal{Y}$: Outcome if treatment were set to t_0.

Causal Estimands:
- Dose-response curve $t_0 \mapsto m(t_0) := \mathbb{E}[Y^{(t_0)}]$;
- Derivative effect curve $t_0 \mapsto \theta(t_0) = m'(t_0) = \frac{d}{dt}\mathbb{E}[Y^{(t_0)}]$.



Issue: When positivity fails at t_0 with $p_{T|\boldsymbol{X}}(t_0|\boldsymbol{x}) = 0$ for units with covariates $\boldsymbol{x}$, $m(t_0)$ and $\theta(t_0)$ need not be *identifiable* without additional assumptions.

Route A: Impose Structural Assumptions

Assume **consistency** ($Y = Y^{(T)}$) and **conditional ignorability** ($Y^{(t)} \perp\!\!\!\perp T | \boldsymbol{X}$), and the following condition:

$$\begin{aligned}\theta(t) &= \frac{d}{dt}\mathbb{E}\left[Y^{(t)}\right] = \mathbb{E}\left[\partial_t \mathbb{E}\left[Y^{(t)} | \boldsymbol{X}\right]\right] \\ &\overset{*}{=} \mathbb{E}\left[\partial_t \mathbb{E}\left[Y^{(t)} | \boldsymbol{X}\right] \middle| T = t\right] \\ &\overset{(**)}{=} \mathbb{E}\left[\partial_t \mathbb{E}(Y | T = t, \boldsymbol{X}) \middle| T = t\right],\end{aligned}$$

where "$\overset{*}{=}$" is the key restriction; (**) uses ignorability and consistency.



By the fundamental theorem of calculus, $m(t_0) = \mathbb{E}\left[Y + \int_T^{t_0} \theta(u)\, du\right]$.

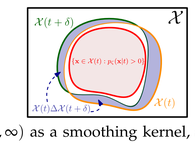
- **Regression Adjustment (RA) Estimator:** Write $\mu(t, \boldsymbol{x}) = \mathbb{E}(Y \mid T = t, \boldsymbol{X} = \boldsymbol{x})$ and $\beta(t, \boldsymbol{x}) = \partial_t \mu(t, \boldsymbol{x})$. Then,

$$\widehat{\theta}_{\mathrm{C,RA}}(t) = \int \widehat{\beta}(t, \boldsymbol{x})\, d\widehat{F}_{\boldsymbol{X}|T}(\boldsymbol{x}|t) \quad \text{and} \quad \widehat{m}_{\mathrm{C,RA}}(t) = \frac{1}{n}\sum_{i=1}^n \left[Y_i + \int_{\tilde{t}=T_i}^{\tilde{t}=t} \widehat{\theta}_{\mathrm{C,RA}}(\tilde{t})\, d\tilde{t}\right].$$

- **Inverse Probability Weighting (IPW) Estimator:**

$$\widehat{\theta}_{\mathrm{C,IPW}}(t) = \frac{1}{nh^2}\sum_{i=1}^n \frac{Y_i \cdot \left(\frac{T_i - t}{h}\right) K\left(\frac{T_i - t}{h}\right) \widehat{p}_\zeta(\boldsymbol{X}_i | t)}{\kappa_2 \cdot \widehat{p}(T_i, \boldsymbol{X}_i)},$$

where $\widehat{p}_\zeta$ estimates the ζ-interior conditional density function $p_\zeta(\boldsymbol{x}|t)$.



- **Doubly Robust (DR) Estimator:** Write $K : \mathbb{R} \to [0, \infty)$ as a smoothing kernel, and $\kappa_2 = \int u^2 K(u)\, du$. Then,

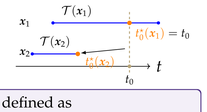
$$\widehat{\theta}_{\mathrm{C,DR}}(t) = \frac{1}{nh^2}\sum_{i=1}^n \frac{\left(\frac{T_i - t}{h}\right) K\left(\frac{T_i - t}{h}\right) \widehat{p}_\zeta(\boldsymbol{X}_i | t)}{\kappa_2 \cdot \widehat{p}(T_i, \boldsymbol{X}_i)} \left[Y_i - \widehat{\mu}(t, \boldsymbol{X}_i) - (T_i - t) \cdot \widehat{\beta}(t, \boldsymbol{X}_i)\right] + \int \widehat{\beta}(t, \boldsymbol{x}) \cdot \widehat{p}_\zeta(\boldsymbol{x}|t)\, d\boldsymbol{x}.$$

Limitation of Route A: The sufficient model for "$\overset{*}{=}$" implicitly assumes a *homogeneous* treatment effect: $Y^{(t)} = \bar{m}(t) + \xi(\boldsymbol{X}) + \epsilon$, with independent, mean-zero ϵ.

Route B: New Estimand Under a Feasible Intervention

For the nonempty and closed conditional support $\mathcal{T}(\boldsymbol{x})$ of $p_{T|\boldsymbol{X}}(t|\boldsymbol{x})$, assume a unique projection almost surely:

$$t_0^\star(\boldsymbol{x}) := \arg\min_{t \in \mathcal{T}(\boldsymbol{x})} |t - t_0|.$$



The *Nearest feasible treatment policy (NFTP)* estimand is defined as

$$\bar{m}_0(t_0) := \mathbb{E}\left[Y^{(t_0^\star)}\right] \overset{(\blacktriangle)}{=} \int_{\mathcal{X}(t_0)} \mu(t_0, \boldsymbol{x})\, d\mathrm{P}_{\boldsymbol{X}}(\boldsymbol{x}) + \int_{\mathcal{X}\setminus\mathcal{X}(t_0)} \mu(t_0^\star(\boldsymbol{x}), \boldsymbol{x})\, d\mathrm{P}_{\boldsymbol{X}}(\boldsymbol{x})$$

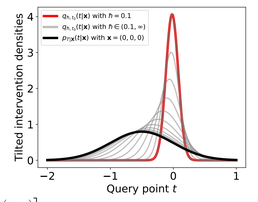
- $Y^{(t_0^\star)}$: potential outcome under the dynamic treatment regime $\boldsymbol{x} \mapsto t_0^\star(\boldsymbol{x})$.
- ($\blacktriangle$): identification without global positivity, and $\mathcal{X}(t_0)$: the support of $p_{\boldsymbol{X}|T}(\boldsymbol{x}|t_0)$.
- $\bar{m}_0(t_0)$ equals the classical dose-response curve $t_0 \mapsto m(t_0)$ under positivity.

Stochastic Tilted Intervention to NFTP

Approach the NFTP intervention through a positivity-lean point-targeted (PLPT) tilted intervention:

$$q_{h,t_0}(t|\boldsymbol{x}) := \frac{\eta(t; t_0, h)\, p_{T|\boldsymbol{X}}(t|\boldsymbol{x})}{\int_{\mathcal{T}} \eta(u; t_0, h)\, p_{T|\boldsymbol{X}}(u|\boldsymbol{x})\, du}.$$

- $t \mapsto \eta(t; t_0, h) \propto \exp\left[-\left|\frac{t - t_0}{h}\right|^\gamma\right]$ for some $\gamma > 0$;
- $h > 0$ governs concentration around t_0.
- $Q_{h,t_0}(\cdot \mid \boldsymbol{x}) \Rightarrow \delta_{t_0^\star(\boldsymbol{x})}(\cdot)$ as $h \to 0$.



Tilted Intervention Estimand: $t_0 \mapsto \bar{m}_h(t_0) := \mathbb{E}\left[Y^{(q_{h,t_0})}\right]$.
- $Y^{(q_{h,t_0})}$: the outcome under treatment drawn from $q_{h,t_0}(t|\boldsymbol{x})$ for each unit $\boldsymbol{X} = \boldsymbol{x}$.
- $\bar{m}_h(t_0) \to \bar{m}_0(t_0)$ recovers the NFTP estimand as $h \to 0$.

One-Step Estimator Based on the Efficient Influence Function:

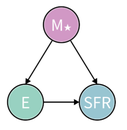
$$\widehat{\bar{m}}_h(t_0) = \frac{1}{n}\sum_{i=1}^n \left\{\frac{\widehat{q}_{h,t_0}(T_i|\boldsymbol{X}_i)}{\widehat{p}_{T|\boldsymbol{X}}(T_i|\boldsymbol{X}_i)}\left[Y_i - \int_{\mathcal{T}} \widehat{\mu}(t_1, \boldsymbol{X}_i)\widehat{q}_{h,t_0}(t_1|\boldsymbol{X}_i)\, dt_1\right] + \int_{\mathcal{T}} \widehat{\mu}(t_1, \boldsymbol{X}_i)\widehat{q}_{h,t_0}(t_1|\boldsymbol{X}_i)\, dt_1\right\}.$$

Asymptotic Normality: Under regularity conditions with $n \to \infty$ and $\hbar \to 0$,

$$\sqrt{\frac{n}{V_{\mathrm{eff},\bar{m}_\hbar}(t_0)}}\left[\widehat{\bar{m}}_\hbar(t_0) - \bar{m}_0(t_0)\right] \xrightarrow{d} \mathcal{N}(0, 1).$$

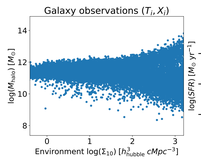
"Nature vs. Nurture" in Galaxy Evolution

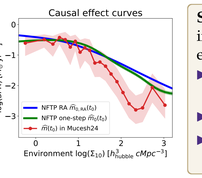
Question: How does local environment ("nurture") affect a galaxy's star formation rate (SFR) after adjustment for its halo mass ("nature")?





- **Data:** 18,629 galaxies with $M_\star > 10^9 M_\odot$ from IllustrisTNG (TNG100), with redshift $z \approx 0$–3.
- **Treatment:** 10-th nearest neighbor density.
- **Outcome:** Instantaneous SFR for each galaxy.
- **Confounder:** Halo mass, $\log(M_{\mathrm{halo}})$.





Scientific finding: After adjusting for halo mass (*nature*), the environmental effect (*nurture*)
- is strongly negative at low redshift,
- weakens for $z > 0.5$, and
- shows no clear positive reversal at high redshift.

